

From CLMI.X to CLMIX-AI: A Machine Learning–Based Upgrade of the Cone Location and Magnitude Index Expanded to Detect Keratoconus Suspects

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Purpose: To enhance the CLMI.X index for detecting keratoconus suspect (KCS) cases by incorporating advanced machine learning (ML) algorithms.

Design: Development and validation of a ML diagnostic algorithm.

Methods: This study included 352 eyes divided into 3 categories: normal (NL) (n = 133), KCS (n = 77), and keratoconus (KC) (n = 142). Imaging was obtained using the Galilei dual Scheimpflug–Placido system. Eleven variables, identical to those used in the original CLMI.X, were employed to train and test multiple ML models, including logistic regression (LR) and more advanced algorithms, for classification of corneas into NL, KCS, and KC. SHapley Additive exPlanations (SHAP) analysis was performed to identify influential variables for differentiating KCS from NL eyes, both for the full KCS group and a subgroup of topographically and tomographically borderline cases.

Results: The original CLMI.X demonstrated high sensitivity (100%) and specificity (99%) for detecting KC but low sensitivity (4%) for KCS. Training the CLMI.X-AI using LR on 3 classes improved KCS sensitivity to 68%. Incorporating advanced artificial intelligence algorithms further increased KCS sensitivity to 75%, with an overall accuracy of 89%. SHapley Additive exPlanations analysis identified the posterior axial, posterior instantaneous, and anterior axial curvature maps as the most influential variables, alongside zonal pachymetry and posterior elevation. For topographically and tomographically NL fellow eyes, SHAP analysis revealed significant influence for zonal pachymetry.

Conclusions: The CLMI.X-AI demonstrates substantially improved performance in detecting KCS compared with the original CLMI.X. Its ML enhanced decision-making and 3-class training make it a more clinically relevant and robust tool for early KC detection.

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Despite recent advances in corneal imaging, detecting keratoconus (KC) in its early stages remains challenging.¹ In 2008, a group of researchers introduced the cone location and magnitude index (CLMI) to detect KC based on anterior axial and tangential curvature maps. The index was calculated based on their algorithm that scans the mapped area searching for the steepest-area local region, then compares it to the corresponding region 180° away, across the corneal center, to check for corneal asymmetry, the hallmark of KC.²

They showed that the CLMI calculated from the axial anterior curvature was the most efficient variable in discerning between normal (NL) and keratoconic corneas.² When compared with keratometric parameters, wavefront aberrations, and pachymetry, the CLMI had the highest sensitivity (89%), specificity (94%), and accuracy (92%) for detection of KC.³

In 2013, the same group extended the CLMI to include axial and tangential maps for both the anterior and posterior curvatures, posterior elevation map, and pachymetry.⁴ The upgraded index was named CLMI.X. When tested and validated for discriminating KC from NL corneas, the CLMI.X outperformed the CLMI and achieved a sensitivity of 99.4% and a specificity of 99.6%, and it is currently widely used. However, despite its great accuracy for discriminating KC, the CLMI.X was not specifically designed to identify KC suspect (KCS) corneas, an important yet clinically challenging entity to detect. Moreover, the algorithm used for its implementation was based on logistic regression (LR), the standard model at the time, while more advanced contemporary machine learning (ML) algorithms have shown to perform better at capturing complex relationships in data.

In this work, we aim to introduce the CLMI.X-AI, an upgraded version of the CLMI.X that, relying on the same variables, is trained using state-of-the-art ML algorithms and designed to identify not only KC and NL corneas, but KCS corneas as well.

Methods

This study was conducted at the Cornea and Refractive Surgery Division of the American University of Beirut Medical Center, Beirut, Lebanon. It adhered to the tenets of the Declaration of Helsinki and was approved by the Institutional Review Board of the American University of Beirut (IRB ID: BIO-2019-0399). The need for informed consent was waived by the Institutional Review Board given the retrospective nature of the study.

Subjects

Patients were divided into 3 groups as determined by their clinical examination and corneal imaging: (1) patients with KC in both eyes; (2) KCS patients; and (3) patients with NL corneas who served as the control group. All patients were imaged using the Galilei dual Scheimpflug–Placido topography or tomography system (Ziemer Group AG) as part of their screening for laser vision correction procedures or during their follow-up visits for KC. Patients were excluded if they had a history of ocular or corneal surgery (e.g., intrastromal ring insertion, laser vision correction, corneal collagen cross-linking, etc.), an inadequate image quality, contact lens wear <2 weeks before the image acquisition, as well as the presence of any retinal disease or corneal scarring. Only 1 eye per patient was included in the study. Apart from a few patients, the majority of the cases included in this study were also utilized in our previous studies.⁵

Patients included in the control group were individuals presenting for laser vision correction, with no family history of KC, NL slit lamp examination and corneal imaging, and who underwent LASIK with no evidence of ectasia in any eye at least for 3 years postoperatively.

Keratoconus was diagnosed based on the evaluation of tomographic maps of anterior and posterior surfaces. The diagnosis consisted of having anterior curvature asymmetry, defined by either 1 major or 2 minor criteria, along with posterior surface abnormality. Major criteria included KC prediction index >30%, an inferior-superior value of >2.38 diopters (D), and >20° of skewed radial axis. Minor criteria included a KC prediction index >18.5% but <30%, an asymmetric bowtie pattern, an inferior-superior value between 1.40 D and 2.30 D, and ≥ 1.00 D of inferior steepening.⁶ Posterior surface abnormality was defined as a highest posterior elevation point >18 μm in the 3-mm central zone or >22.5 μm in the 5-mm central zone.⁷

Keratoconus suspect corneas were identified as the fellow eyes of patients with highly asymmetric KC. While these corneas did not meet the pre-established criteria for KC diagnosis, their contralateral eye had been diagnosed with the disease. Keratoconus suspect corneas included tomographically and topographically NL fellow eyes and tomographically and topographically borderline fellow eyes, classified based on both objective criteria (e.g., KC Probability Index, Asphericity Asymmetry Index) and subjective assessment, as detailed in our previous work.⁵

Variables

Eleven variables were utilized to train the models, as outlined in Table 1, and were identical to those used for the CLMI.X. The variables were collected from the Galilei CLMI.X report and

Table 1. Variables Used for the CLMIX and the CLMIX-AI, Calculated Based on the Corneal Curvature, Elevation, and Pachymetry Maps

Parameter	Description
CLMlaa	The magnitude of CLMI calculated on the anterior axial map (D)
ZKmax-aa	The zonal average value of C1, the steepest region on the anterior axial map (D)
CLMI Paxial	The magnitude of CLMI calculated on the posterior axial map (D)
ZK Paxial	The zonal average value of C1, the steepest region on the posterior axial map (D)
CLMI Inst	The magnitude of CLMI calculated on the anterior instantaneous (tangential) map (D)
ZKmax Inst	The zonal average value of C1, the steepest region on the anterior instantaneous (tangential) map (D)
CLMI Pinst	The magnitude of CLMI calculated on the posterior instantaneous (tangential) map (D)
ZKmax Pinst	The zonal average value of C1, the steepest region on the posterior instantaneous (tangential) map (D)
FLMI Pach	The difference C1–C2 calculated on the pachymetry map (μm). The distance between C1 and C2 is ≥ 2 mm
ZTPach	The average value of C1, the thinnest region on the pachymetry map (μm)
PE	The average value of the highest region above the posterior best-fit sphere map (μm)

CLMI = cone location and magnitude index; D = diopters; FLMI = flat location and magnitude index; ZTPach = zonal thinnest pachymetry.

included the magnitudes of the CLMI and the zonal averages of the steepest region on both axial and tangential curvature maps for both anterior and posterior surfaces, the average value of the posterior elevation in the highest region above the best-fit sphere, the average value of the corneal thickness in the thinnest region, and the difference of corneal thickness between the thinnest region and the diametrically opposite region. The search routine of the CLMI algorithm is shown Figure 1.

ML Workflow

The data were initially examined to ensure absence of missing values or inconsistencies. Next, all variables were standard-scaled by subtracting the mean and dividing by the standard deviation, an important step to mitigate the influence of differences in measurement units and magnitudes, during the training phase.

Recognizing that each ML algorithm has specific strengths and limitations, several algorithms were trained for the classification task, including the LR, Support Vector Machine, Decision Tree, Naïve Bayes, K-nearest Neighbor, Random Forest, XGBoost, and CatBoost.

The patients were then split into training set (80%), providing data needed to learn underlying patterns by the model, and testing set (20%), used to assess the performance of the model on new data. To address class imbalance in the training set, a synthetic minority oversampling technique⁸ was used. It works by interpolating instances of the minority class to alleviate the bias toward the majority class, ensuring thus less overfitted and more generalizable models.

To efficiently identify the optimal set of hyperparameters for each algorithm and reduce computational costs, a random search grid was used during the training phase. Finally, a fivefold cross-validation was applied to provide a more robust assessment of the

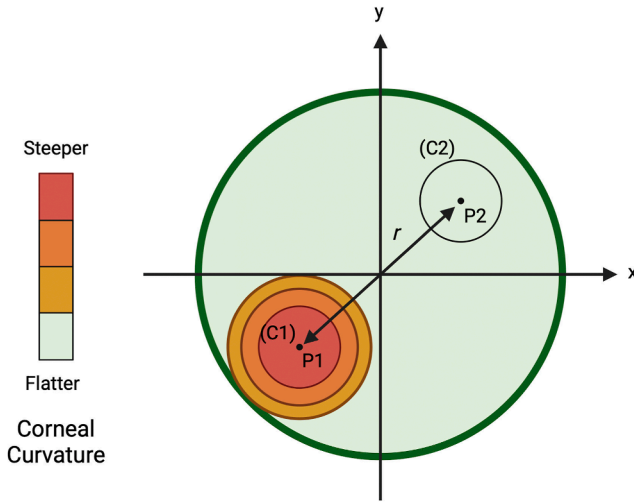


Figure 1. Confusion Matrix of the original CLMIX showing poor sensitivity for detection of KCS. The algorithm identifies a 2-mm diameter circle (C1), including the steepest region of the map. The center of this circle is denoted as P1 and is located at a distance r from the center of the map. P2 is defined as the point located 180° from P1, and a second 2-mm diameter circle (C2) is centered at P2. This approach is applied to anterior and posterior maps and adapted for pachymetry map analysis to compute the CLMIX. CLMI = cone location magnitude index.

model's performance. All steps were implemented using the Imbalanced learn pipeline and the Scikit-learn library on Python 3.7.

When evaluated, the models were penalized for underestimating the risk of KCS⁹ and were assessed based on their sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), F1-score, and accuracy.

Experiments

As aforementioned, the goal of this work was to optimize the decisional algorithm of the CLMIX by implementing 2 steps: (1) using an artificial intelligence-enhanced model rather the LR model; and (2) including KCS cases as a separate class in the training process.

To do so, 2 experiments were designed. In experiment 1, an LR model was trained to classify corneas into 1 of the 3 classes: NL, KCS, and KC and compared with the CLMIX (trained on NL and

KC only). In experiment 2, several AI algorithms were trained to classify corneas into 1 of the 3 classes (NL, KCS, and KC), and then compared with the LR model. Moreover, SHapley Additive exPlanations (SHAP)¹⁰ values were computed to identify the most important variables influencing the model's decision-making between NL and KCS eyes. This analysis was performed twice, once using the entire KCS group, then using only the topographically and tomographically borderline eyes of the KCS group. This approach was adopted for a deeper understanding of the model's behavior across different nuances of abnormalities in the context of suspect corneas.

Statistical Analysis

Categorical variables were described using frequencies and proportions, and compared using chi-square test. Continuous variables were described using means, standard deviations, and ranges, and their normality was evaluated using the Shapiro–Wilk test. To test for differences among groups, analysis of variance was utilized for normally distributed variables, while the Kruskal–Wallis test was employed for non-normally distributed variables. A significance level of $P < 0.05$ was adopted for all analyses. Statistical computations were conducted using IBM SPSS Statistics software version 26.

Results

The study included a total of 352 eyes of 352 patients divided as follows: 133 NL control corneas, 142 corneas with definite KC, and 77 KCS corneas. The vast majority (96%) of the KC eyes were grade 1 or 2 based on the Amsler–Krumeich classification. Patient demographics and characteristics are summarized in Table 2. There was no statistically significant difference between groups except for the mean keratometry and corneal thinnest point.

Table 3 summarizes the results of best models across different experiments.

The original CLMIX outputs a probability score of KC as follows:⁴

- CLMIX score $\geq 80\%$: KC
- 25% less than or equal to CLMIX score $< 80\%$: KCS
- CLMIX score $< 25\%$: NL

Table 2. Patient Demographics and Ocular Parameters

Parameter	Mean $\pm$ Standard Deviation (Range)			P Value
	NL	KCS	KC	
Number of eyes, n (%)	133 (37.8%)	77 (21.9%)	142 (40.3%)	-
Age	37.22 $\pm$ 12.30 (19; 70)	35.49 $\pm$ 12.54 (21; 78)	36.01 $\pm$ 12.95 (11; 73)	0.399
Sex (% males)	47%	53%	59%	0.145
KMean (D)	43.78 $\pm$ 1.90 (39.01; 48.56)	44.52 $\pm$ 2.44 (39.97; 47.92)	44.71 $\pm$ 2.28 (40.52; 56.87)	0.009
CCT (μ m)	550.24 $\pm$ 30.84 (498.00; 645.00)	514.71 $\pm$ 34.03 (425.00; 577.00)	471.61 $\pm$ 51.73 (323.00; 589.00)	<0.001

CCT = central corneal thickness; D = diopters; KMean = mean keratometry; KC = keratoconus; KCS = keratoconus suspect; NL = normal. Bold values indicate statistical significance (P value < 0.05).

Table 3. Results of the Different Experiments

Experiment	Algorithm	Training Classes	Class	Accuracy (%)	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)	F1-Score
Original CLMIX	Logistic regression	NL vs. KC	NL	79.7	100	67	65	100	0.78
			KCS		4	100	100	79	0.08
			KC		100	99	99	100	0.99
Exp (1)	Logistic regression	NL vs. KCS vs. KC	NL	85.0	85	88	81	91	0.80
			KCS		68	90	63	89	0.66
			KC		93	98	97	95	0.95
Exp (2)	CatBoost	NL vs. KCS vs. KC	NL	89.0	84	91	85	90	0.85
			KCS		75	92	69	91	0.73
			KC		99	97	96	99	0.97

KC = keratoconus; KCS = keratoconus suspect; NL = normal; NPV = negative predictive value; PPV = positive predictive value. Bold values indicate statistical significance (P value < 0.05).

When applied to this dataset, the original CLMIX (Fig 2) achieved a sensitivity of 100% and specificity of 99% for detecting KC, but the sensitivity was only 4% for KCS, with most misclassified corneas being labeled as NL. Figure 3A and B show the corneal maps of 2 different KCS patients who were misclassified by the CLMIX as NL. As shown in both cases, the CLMIX correctly identifies the area of asymmetric inferior steepening, focal thinning, and highest posterior elevation; however, still classifies these patients as NL (CLMIX score of 0% and 14%, respectively).

Training the model on 3 classes (NL, KCS, and KC) with LR (Experiment 1, Fig 4A) significantly improved KCS detection, with 68% sensitivity, 90% specificity, 63% PPV, and 89% NPV, with a drop in KC sensitivity to 93%.

In contrast, employing more advanced AI algorithms (i.e., CatBoost in Experiment 2, Fig 4B) further enhanced detection for both KCS and KC: 75% sensitivity, 92% specificity, 69% PPV, and 91% NPV for KCS, with 99% sensitivity and 97% specificity for KC. Notably, unlike

the 3-class LR model (Experiment 1), no misclassified KC corneas were labeled as NL using the CatBoost-based CLMIX-AI. Notably, when the patients presented in Figure 3 were tested using the CLMIX-AI, both were correctly classified as suspects.

The SHAP analysis for NL versus KCS (Fig 5A) showed that the variables that most influenced the decision-making of the model were the magnitudes of the CLMI calculated on the posterior axial map, the zonal average of pachymetry in the thinnest region, the magnitudes of the CLMI on posterior and anterior instantaneous maps, as well as the zonal posterior elevation in the highest region. When the analysis was repeated after excluding topographically and tomographically NL fellow eyes from the KCS group, the SHAP analysis showed that the zonal pachymetry of the thinnest region dropped from the being second to fourth most influential variable (Fig 5B).

Discussion

The CLMIX is a robust index that analyzes multiple topographic and tomographic parameters from the anterior and posterior corneal surfaces to differentiate KC from NL eyes.⁴ However, its current formula does not include identification of KCS; a clinically challenging yet important entity to detect early, especially in refractive surgery candidates. Therefore, enhancing the CLMIX by enabling the detection of suspect cases would be of greater value to clinicians by facilitating screening and minimizing risks associated with refractive procedures.^{11,12} In this study, we aimed to retrain the core parameters of the CLMIX using ML models and test them on 3 categories of eyes: NL, KC, and most importantly, KCS, which are the fellow eyes of very asymmetric ectasia patients, and typically are either tomographically and topographically NL, or borderline.

In line with the findings from the literature,⁴ our results showed that the original CLMIX achieved excellent sensitivity and specificity for detecting KC. However, a very low sensitivity was achieved for the KCS. As illustrated in Figure 3, although the original CLMIX model successfully pinpointed zones of abnormalities on

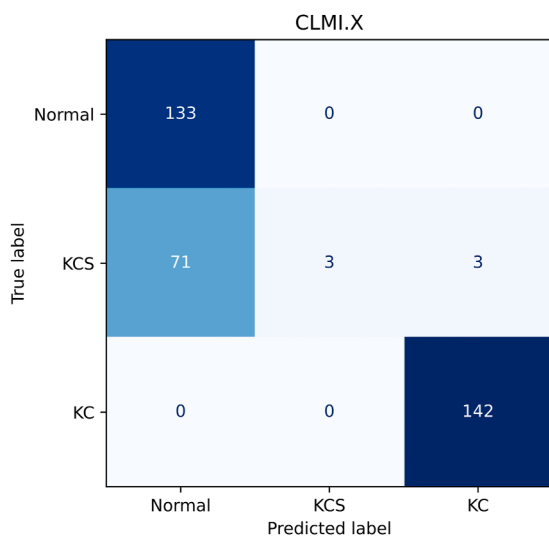
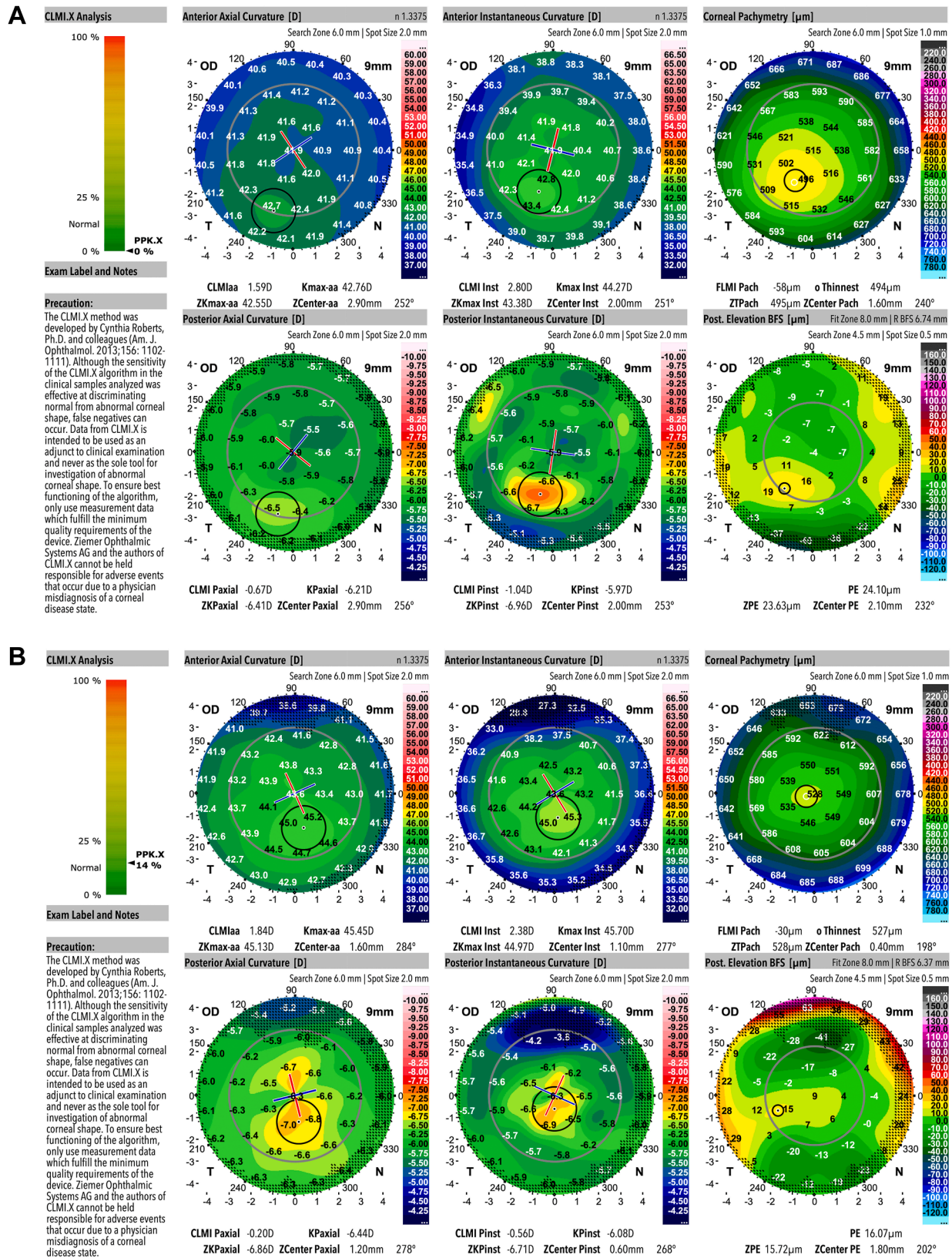


Figure 2. Confusion matrix of the original CLMIX. KC = keratoconus; KCS = keratoconus suspect.



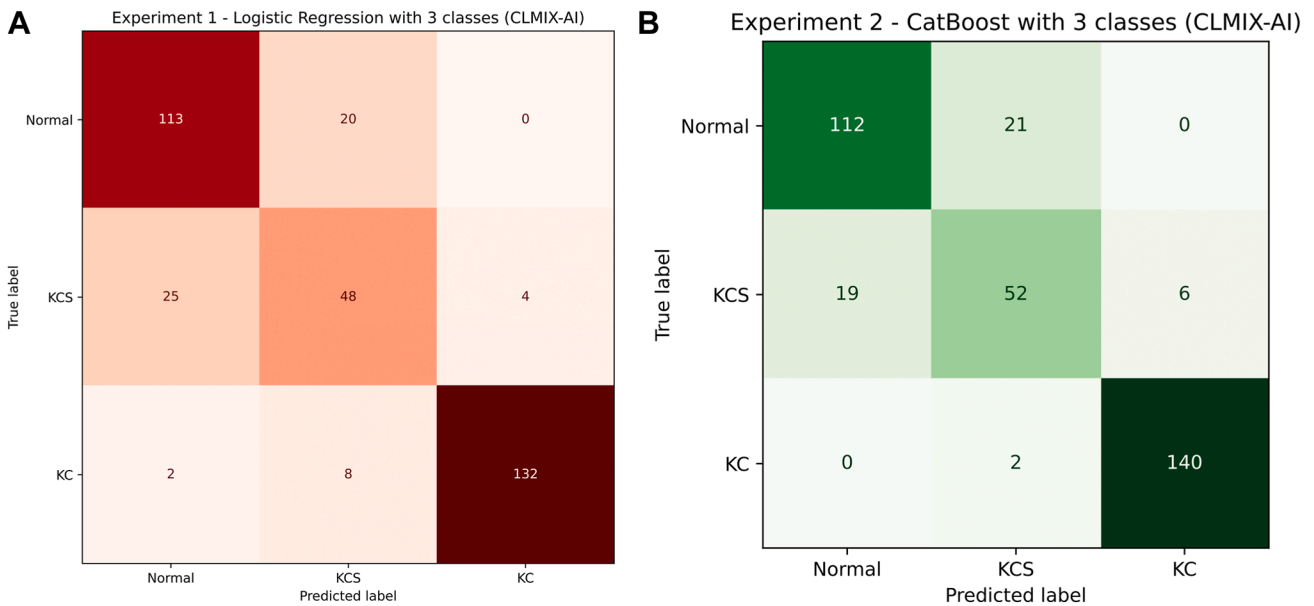


Figure 4. Confusion matrices of (A) the CLMIX trained on 3 classes (normal, KCS, and KC) using logistic regression (Experiment 1) and (B) the CLMIX-AI trained on 3 classes using artificial intelligence algorithms (CatBoost, Experiment 2). The CLMIX-AI shows higher sensitivity for detection of KCS and does not misclassify of KC as normal. KC = keratoconus; KCS = keratoconus suspect.

corneal maps (e.g., steepening, thinning, and elevation), it still misclassified these corneas as NL. This suggests that the low sensitivity for detecting KCS corneas is not attributable to the input variables but rather to limitations in the algorithm's final decision-making process.

Two potential reasons may explain this: the model may lack familiarity with the milder abnormalities in KCS corneas, as they are less pronounced than those observed in KC corneas or the complexity of the relationships among variables may exceed the capacity of LR to capture effectively. These challenges underscore the importance of the 2 conducted experiments in identifying and addressing the factors impeding accurate predictions. In fact, the first experiment demonstrated that introducing the KCS corneas into the model training, as a separate class, significantly improved its sensitivity for detecting these cases. The second experiment further confirmed that upgrading the decisional algorithm to an artificial intelligence-based approach enhanced performance across all classes.

The SHAP analyses revealed that the most influential variable for the model for NL classes NL and suspects was asymmetry of axial curvature on the posterior surface. In

addition, after excluding the tomographically and topographically NL fellow eyes from the KCS group, leaving only the tomographically and topographically borderline fellow eyes, corneal pachymetry dropped from the second to the fourth most influential variable for differentiating between KCS and NL corneas. This finding suggests that decreasing pachymetry is likely one of the earliest measurable parameters that is present and characteristic of these seemingly NL corneas that might be potentially at risk of progression. Previous work by our group¹³ showed differences in the mechanical properties of some NL and tomographically and topographically NL fellow eyes corneas using the Corvis ST (Oculus Optikgeräte GmbH). Although this remains a topic of debate, the results suggest that tomographically and topographically NL fellow eyes, despite lacking overt abnormalities on imaging, may represent the earliest and most challenging cases of KCS. Therefore, including these cases in this study is thus crucial for exposing the model to patterns indicative of the earliest stages of KCS.

The SHAP findings further suggest that the earliest changes in KCS may be associated with alterations in

instantaneous (tangential) map (D); CLMI_Pinst = the magnitude of CLMI calculated on the posterior instantaneous (tangential) map (D); CLMIaa = the magnitude of CLMI calculated on the anterior axial map (D); FLMI_Pach = the difference C1–C2 calculated on the pachymetry map (μm). The distance between C1 and C2 is ≥ 2 mm; OD = right eye; KCS = keratoconus suspect; PE = the average value of the highest region above the posterior best-fit sphere map (μm); ZKmax_aa = the zonal average value of C1, the steepest region on the anterior axial map (D); ZKmax_Inst = the zonal average value of C1, the steepest region on the anterior instantaneous (tangential) map (D); ZKPaxial = the zonal average value of C1, the steepest region on the posterior axial map (D); ZKPinst (or ZKmaxPinst) = the zonal average value of C1, the steepest region on the Posterior Instantaneous(tangential) map (D); ZPE = the average value of the highest region above the Posterior Best-Fit Sphere map (μm); ZTPach = zonal thinnest pachymetry.

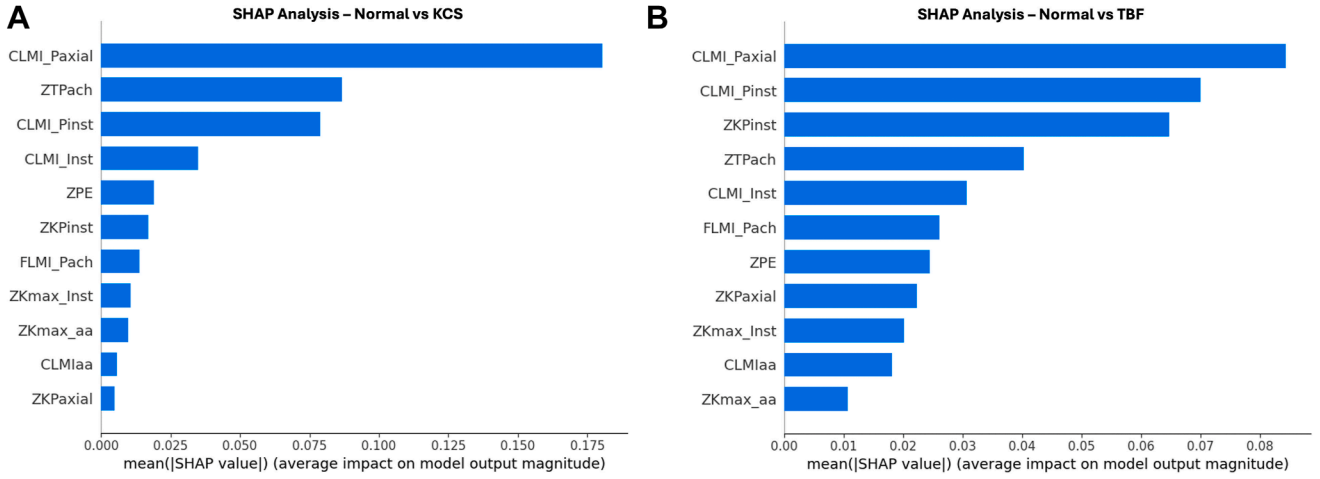


Figure 5. Feature importance for the CLMIX-AI obtained by SHAP analysis from (A) normal vs. KCS (including topographically and tomographically normal and TBF) and (B) normal vs. topographically and TBF eyes only. Zonal pachymetry drops from being second to forth most influential variable after excluding topographically and tomographically normal fellow eyes from the analysis. CLMI = cone location magnitude index; CLMI_Inst = the magnitude of CLMI calculated on the anterior instantaneous (tangential) map (D); FCLMI_Pinst = the magnitude of CLMI calculated on the posterior instantaneous (tangential) map (D); CLMIaa = the magnitude of CLMI calculated on the anterior axial map (D); KC = keratoconus; KCS = keratoconus suspect; LMI_Pach = the difference C1–C2 calculated on the pachymetry map (μm). The distance between C1 and C2 is ≥ 2 mm; SHAP = SHapley Additive exPlanations; TBF = topographically and tomographically borderline fellow eyes; ZKmax_aa = the zonal average value of C1, the steepest region on the anterior axial map (D); ZKmax_Inst = the zonal average value of C1, the steepest region on the anterior instantaneous (tangential) map (D); ZKPaxial = the zonal average value of C1, the steepest region on the posterior axial map (D); ZKPinst (or ZKmaxPinst) = the zonal average value of C1, the steepest region on the Posterior Instantaneous (tangential) map (D); ZPE = the average value of the highest region above the Posterior Best-Fit Sphere map (μm); ZTPach = zonal thinnest pachymetry.

corneal pachymetry. This observation aligns with existing literature, which posits that the KC biomechanical decompensation cycle begins with focal biomechanical weakening, leading to stress redistribution and subsequent focal thinning.¹⁴ Focal steepening is hypothesized to occur as a secondary change. While the role of Corvis ST in assessing these asymmetric changes has its limitations,^{15,16} this hypothesis warrants further investigation with more promising biomechanical tools capable of assessing asymmetry, such as Brillouin microscopy.

When compared with other studies in the literature, the performance of the CLMIX-AI for detection of KCS, despite being based on imaging solely, is comparable to that achieved using biomechanical parameters. In fact, in 2022, our group compared the ability of the tomographic and biomechanical index (TBI), the Corvis biomechanical index, and the stiffness parameter at applanation 1 to differentiate suspect corneas from NL eyes.¹³ The results showed that TBI had the best sensitivity (74.5%) with a PPV of 81.4% and a NPV of 88.5% for detecting KCS. Furthermore, Kataria et al¹⁷ analyzed the discriminative ability of the TBI, Corvis biomechanical index, stiffness parameter at applanation 1, and the Belin–Ambrosio deviation index in distinguishing NL eyes from subclinical and mild KC. When differentiating NL eyes from eyes with very asymmetric ectasia but with NL topography and tomography, the TBI achieved the highest accuracy (79.3%)

and sensitivity (82%) while the Belin–Ambrosio deviation index scored 68% in accuracy and 80% in sensitivity. In our study, the sensitivity for detecting KCS was 75% with an overall accuracy of 89%. Chan et al¹⁸ studied in Asian eyes the efficacy of the SCORE analyzer (Bausch + Lomb TechnoLas) that included multiple parameters such as thinnest pachymetry, maximum posterior elevation, and anterior elevation at the thinnest point. The SCORE detected forme fruste KC with a sensitivity of 70.8%, a specificity of 98.1%, a PPV of 89.5%, and an NPV of 93.6%. In addition, and using swept-source OCT, parameters pertaining to anterior and posterior curvature as well as pachymetric data were combined and fed to a ML model; a discriminant analysis showed a sensitivity of 75% and a specificity of 99% in detecting early KC.¹⁹ However, a 2-class analysis was applied in the latter study (KC vs. NL, KCS vs. NL), compared to a 3-class analysis in our current study (KC vs. KCS vs. NL).

Although pairwise analysis of 2 classes is known to yield better performance metrics, a 3-class analysis is more robust because it provides a more realistic simulation of clinical screening, where any developed algorithm must differentiate among all three categories simultaneously.

There are limitations to our work. First, the study is retrospective. Second, the dataset is not fully balanced across classes, which could impact the model's performance for the minority class. While we used synthetic minority oversampling technique to address this imbalance,

this remains a challenge for clinical datasets due to the scarcity of KCS cases. Third, the dataset originates from a single center which could restrict the diversity of the data. Although we performed cross-validation to ensure the model's scalability, an ideal next step would involve external validation on an independent test set from a different location to confirm the robustness of the model.

This study presents the CLMI.X-AI, an upgraded version of the CLMI.X index, optimized using advanced ML algorithms and incorporating KCS as a distinct training class, providing a more robust and clinically relevant tool for early KC detection. Future work will focus on external validation using diverse datasets and exploring multimodal data integration to enhance diagnostic accuracy further.

Footnotes and Disclosures

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HUMAN SUBJECTS: Human subjects were included in this study. It adhered to the tenets of the Declaration of Helsinki and was approved by

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No animal subjects were used in this study.

Author Contributions:

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Analysis and interpretation: Hammoud, Wehbi, Assaf, Roberts, Awwad

Data collection: Hammoud, Assaf

Overall responsibility: Hammoud, Wehbi, Roberts, Awwad

Abbreviations and Acronyms:

CLMI = cone location and magnitude index; **D** = diopters; **LR** = logistic regression; **KC** = keratoconus; **KCS** = keratoconus suspect; **ML** = machine learning; **NL** = normal; **NPV** = negative predictive value; **PPV** = positive predictive value; **SHAP** = SHapley Additive exPlanations; **TBI** = tomographic and biomechanical index.

Keywords:

Artificial intelligence, Keratoconus, Keratoconus suspects, Refractive surgery screening, Subclinical keratoconus.

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